***** WRITE YOUR NAME AND COHORT BELOW *****

UNIVERSITY OF CALIFORNIA BERKELEY
Haas School of Business

MBA 201A—Economic Analysis for Business Decisions—Fall 2021
Professor Steve Tadelis

MIDTERM EXAM

Instructions: The number in brackets (*e.g.,* [5]) indicates the points for each question. Total: 120 points. You have 120 minutes to complete the exam, so if stay slightly under 1 minute per point, you will have some time to check your work.

Please Write Legibly. Briefly explain your answers (that is, don't just write "yes" or "no" and don't just write down a numerical answer without showing how you derived it). Write only on this exam and not on other sheets of paper. The last page is a blank sheet for your use.

****** Please sign the honor code oath at the bottom of this page. ******

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Honor Code:

I understand that this exam is an individual exercise. I swear on my honor that I have not consulted with another person, made use of notes or other materials on- or off-line, or used more than the 4-function basic calculator functions ($+, -, \times, \div$) during the exam.

Name:_____

Cohort:_____

Short answer questions 1-3 [24 points]

These 3 questions require **short answers** (up to 5 lines/sentences). Use any graphs or calculations that will help your explanation and label graphs clearly.

1. [8] A WSJ article published about two weeks ago titled “Peloton Slashes Price on Exercise Bike, Predicts Slowing Growth,” explains that after raising prices significantly in 2020, the company recently dropped prices on its lower-end bike by 20%. Does this imply that Peloton’s price hike last year was a mistake?

2. [8] A college friend recently developed a transparent facemask that does not fog up. For the first 8 weeks they priced it on Amazon at $25, selling a steady average of 1,000 masks a week and just broke even. To sell more masks, during the past two weeks they cut the price to $20 and succeeded to sell about 1,100 masks a week. Assuming demand is stable, can you conclude whether their profit rose or dropped? What advice would you offer your friend to achieve even higher profits?

3. [8] Your cousin who owns a restaurant is asking you for pricing advice. Given the recent anxiety around the Covid Delta variant, their landlord reduced the rent for the restaurant by 10%. Your cousin shares with you that they are quite certain they should take advantage of this and reduce prices by 5% to increase profits. Would you suggest that they proceed with this choice or not?

MULTI-PART QUESTIONS

Question 4 [30 points]

Cisco Networks is facing a tough decision. Smartly anticipating the recent chip shortage, they already invested $500 Million in building manufacturing capacity in Taiwan, expecting to deliver their first batch of chips to market in January 2022. Given the rapid technological innovation, this capacity will last for 4 years, after which their manufacturing capacity will become obsolete. Because of the low interest rates, Cisco's FP&A group does not discount future funds if the time horizon is less than 5 years. On September 17, 2021, it became clear that they must purchase additional equipment from Applied Materials to fully learn what their production capabilities will be. The additional equipment costs are $300 Million. After this additional equipment is installed, the remaining uncertainty about Cisco's manufacturing costs and capacity will be fully resolved. The R&D team projects that one of three scenarios will materialize by January 2022 after this additional equipment is purchased and installed:

- The worst-case scenario will require an additional manufacturing investment of $1 Billion (beyond the $300 Million) and will result in variable costs of about $10 a chip and a production and sales of 3 million chips a year.
- A more promising scenario will require only an additional $400 Million in manufacturing investment (beyond the $300 Million) and will result in variable costs of about $8 a chip and a production and sales of 5 million chips a year.
- The best-case scenario will require no additional manufacturing investment, variable costs of about $2 a chip, and a production and sales of 10 million chips a year.

The R&D team estimates that the worst-case scenario has a probability of 0.4 and the best-case scenario has a probability of 0.2. Also, for decision making purposes, the company is assuming that regardless of the scenario that materializes, the company will sell the chips at $50 each.

a. [15] With no additional information, and assuming that Cisco is risk neutral with respect to this decision, should Cisco invest the $300 million over the next 3 months to see which scenario is realized? Draw a decision tree as part of your analysis.

b. [5] If Cisco were risk averse, might that change your answer in part (a) above? Why or why not??

c. [10] Applied Materials offers Cisco a buy-back option for the $300 million equipment: if Cisco chooses to abandon their chip production aspirations in Q1 of 2022, before any units are produced but after uncertainty is resolved, then Applied will buy back the equipment for $100 Million. Applied is asking $50 Million for this option. Should Cisco buy the option? What is Cisco's maximum willingness to pay for it?

Question 5 [36 points]

Given the lack of fine food near campus, you co-founded of Haas Fine Foods (HFF) of Berkeley, an importer of superior specialty snacks and ingredients. Among your best-selling items is pršut, a specialty ham from the Istria region of Croatia (pršut is similar to Italian prosciutto).

Weekly demand for pršut, in pounds, is expected to be very stable over the next few years and is given by $p(q) = 150 - \frac{3}{2}q$. HFF's overhead costs (rent, electricity, and checkout staff) come to $1,100 a week, and sales of pršut make up approximately five percent of their revenues. Moreover, HFF's supplier charges $30 per pound of pršut.

a. [16] What price, per pound of pršut, would maximize HFF's profits? How much pršut will be sold at that price, and what are the operating profits from pršut?

After some time, HFF's supplier approaches them with an opportunity to submit a bulk order. For a commitment of $67,600 (or $67,600/52 = $1300 per week), they will send HFF a constant amount of 60 pounds of pršut a week for the next year. Any additional amounts can be purchased at $30 dollars per pound as before. For now, assume that unsold pršut is wasted and must be thrown away at the end of each week.

b. [10] Your partner and co-founder, without consulting you, signs HFF up for the annual bulk order contract. Does this change the optimal price? If so, what price would you set?

c. [5] Did your partner do the right thing for the business? That is, did profits increase?

d. [10] After your partner accepted the bulk order, a local chef offers to purchase any unsold pounds of pršut for $18 each. Now, what price should you set?

Question 6 [30 points]

After Steve finally managed to overcome his chocolate addiction, he had 80 bars of excellent Trader Joes chocolate left in his office and decided to give them to you for free. You learn that you can set up a one-time stand in the Hass courtyard and sell the chocolate there. You know that demand from folks in the courtyard is given by $q_H = 200 - 20p$. You also learn that you can take Bart to San Francisco and sell chocolate outside the exclusive Chocolate Lover's Convention, where demand is given by $q_L = 200 - 10p$. You can choose to sell in one or both markets, and it is safe to assume that there will be no arbitrage across these markets. The roundtrip on Bart costs $10 and assume that your opportunity cost of time is zero.

a. [20] Which markets do you choose to sell in, at what price(s), and what are your profits?

b. [10] You learn that Bart is shut down and to go to SF you need to take an Uber or Lyft. What is your maximum wiliness to pay for the round trip?